

Text request

Return amount
in May?

lookup "return amount"

1 Match name

name or alias contains it

store return amount ✓
store return rate —
store revenue —

2 Compare versions

did its tables change?

	last check	now
store_returns	14	15
date_dim	3	3

changed: recheck needed

3 Reuse or check

look up each condition's result

one row per return	found at 15: reuse	✓
one date per date key	found at 3: reuse	✓
no lost dates	none at 15: check, store	✓

4 Return

to the agent, if valid

revision 2, valid
store_returns
SUM(sr_return_amt)
sr_status='completed'

Reuse of check results

Each check result is kept under its condition and the version of the tables the check read. Every definition and agent that needs the same condition on the same version reuses it; only a new version needs a new check.

condition	table version		stored by	reused in
one row per return	store_returns 15	✓	return rate	3
one date per date key	date_dim 3	✓	an earlier check	3 6
no lost dates	store_returns 15	✓		3
one row per return	store_returns 16	✓		6
no lost dates	store_returns 16	✓		6

SQL request

```
SELECT SUM(sr_return_amt)
FROM store_returns
JOIN date_dim ON ...
WHERE sr_status = 'completed'
AND d_moy = 5 AND ...
```

declares: return amount, revision 2

5 Check the declaration

can revision 2 be used?

✓ exists, not invalidated
✓ still the latest revision
✓ SQL keeps the status filter
✗ revision 1: outdated, rejected

6 Same-snapshot validation

conditions hold on the query's snapshot

a write committed: store_returns now at 16

one row per return	none at 16: check, store	✓
one date per date key	found at 3: reuse	✓
no lost dates	none at 16: check, store	✓

7 Execute

on the same snapshot

run the query
answer 3,294,349.93
a failed condition
rejects the query